Parity and invariants

High school · reasoning · Progressive exercises · 01

Exercise 1. Start at 7 and add or subtract 4 each move. Can you reach 20? What about 23?

Exercise 2. Five numbers, each 1 or -1, have product -1. Prove that changing two signs per move cannot make all five positive.

Exercise 3. Replace two integers a and b by $a + b$. What invariant does this preserve?

Exercise 4. Why need two states of the same parity not be connected by allowed moves? Give an example.